Ch 4 Classification

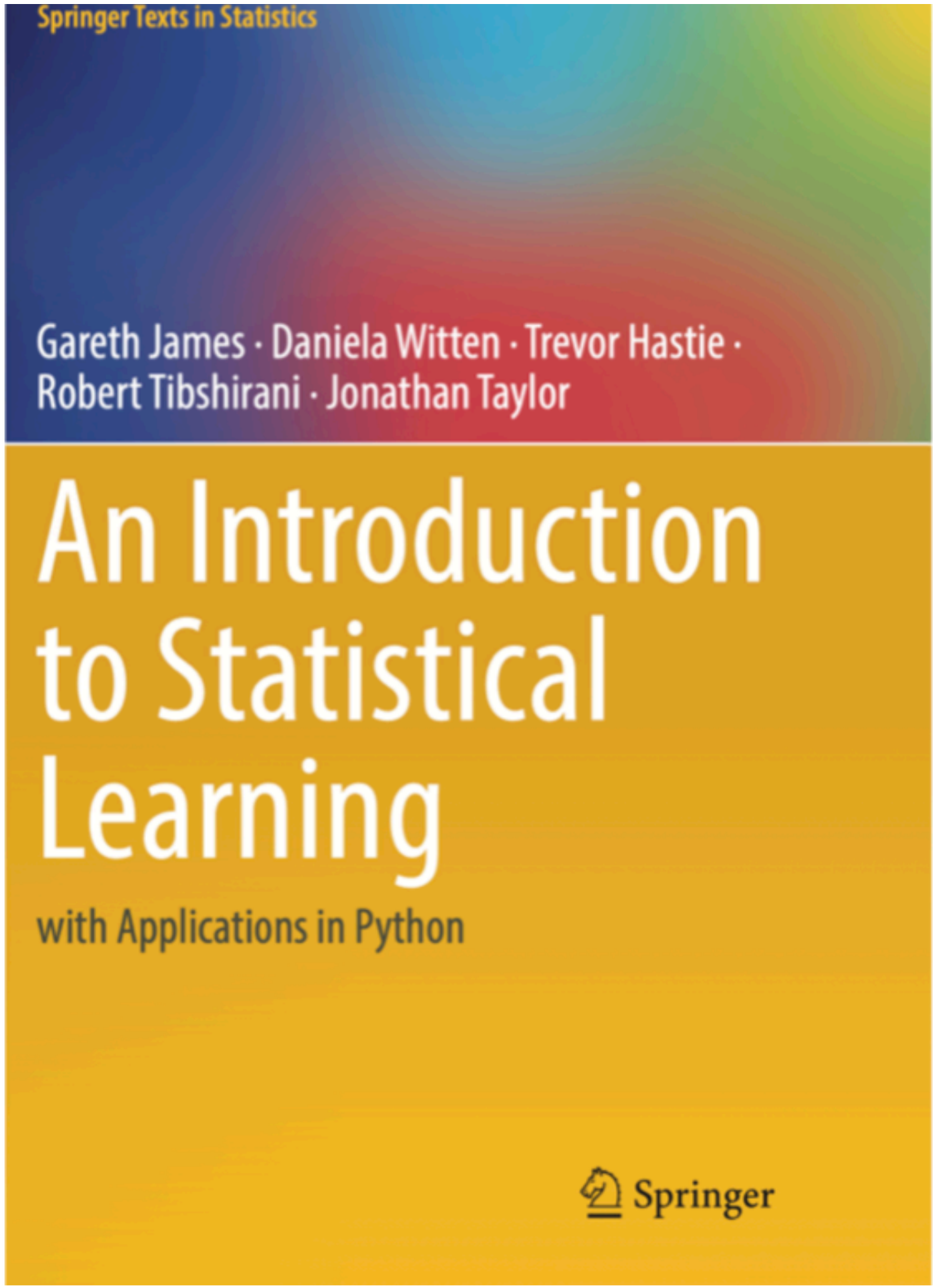
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Slides modified from https://www.statlearning.com/

October 20, 2023

4.1

An overview of Classification



https://www.statlearning.com/

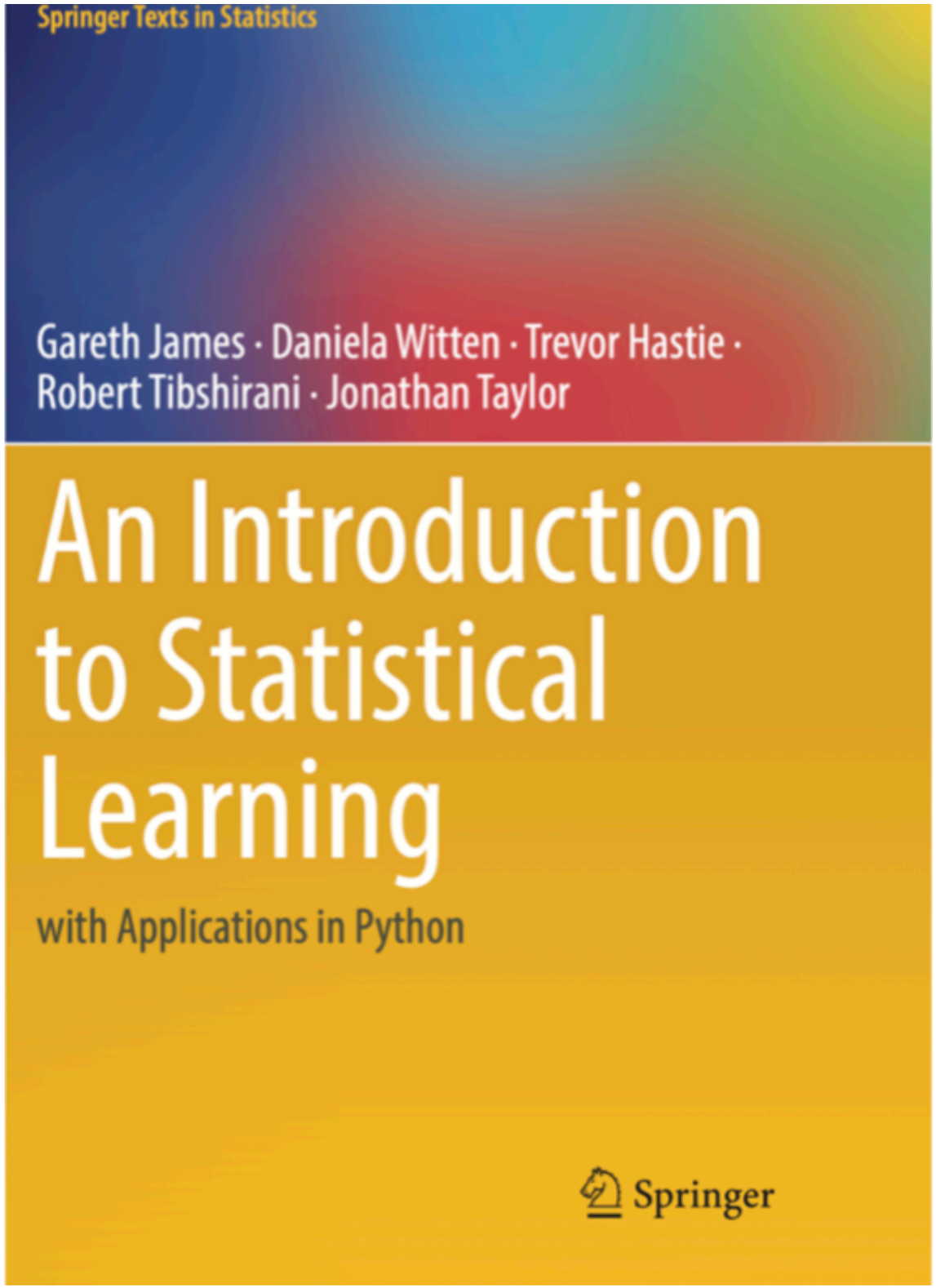
Classification

- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:
  `eye color` $\in \{$`brown`, `blue`, `green`$\}$
  `email` $\in \{$`spam`, `ham`$\}$.
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y; i.e. $C(X) \in \mathcal{C}$.
- Often we are more interested in estimating the *probabilities* that X belongs to each category in $\mathcal{C}$.

  For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

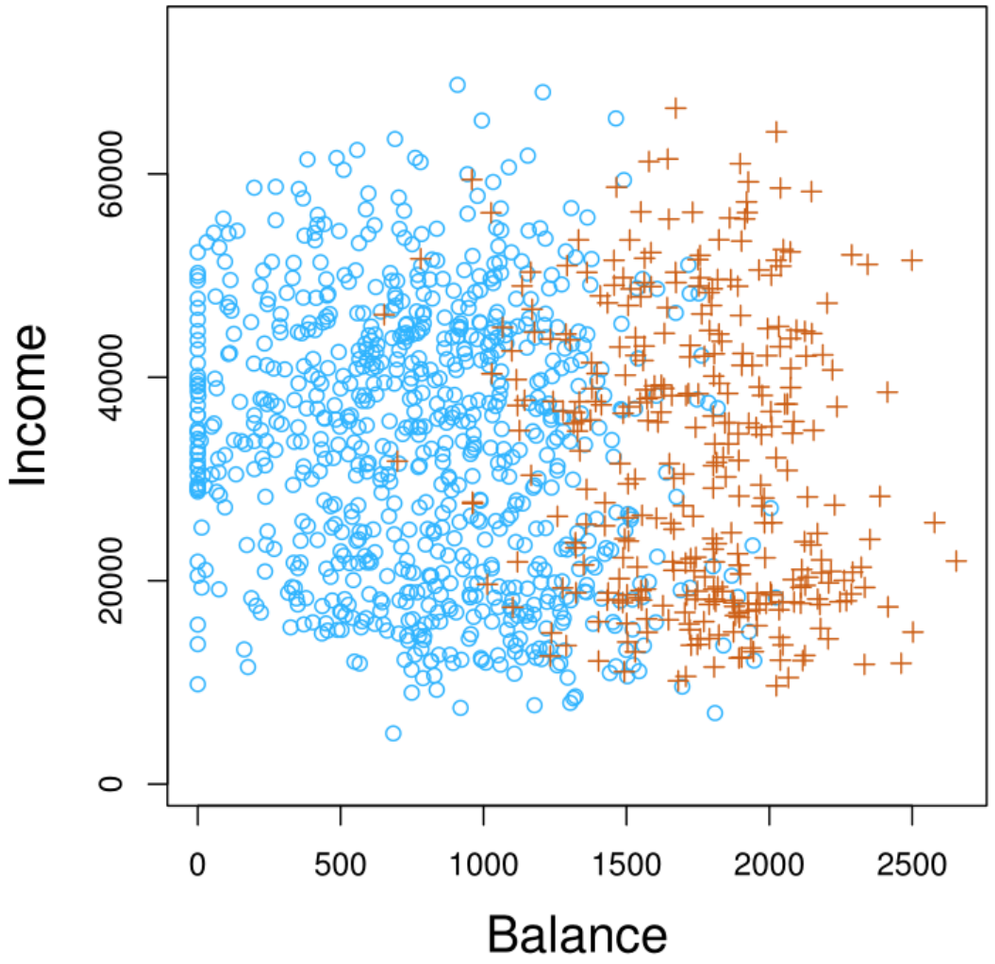
4.2

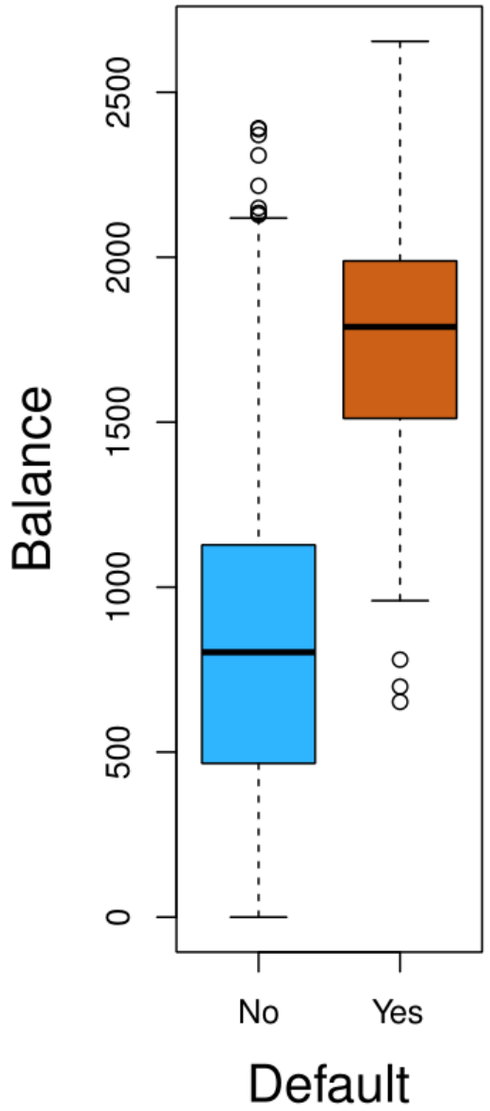
Why not linear regression?

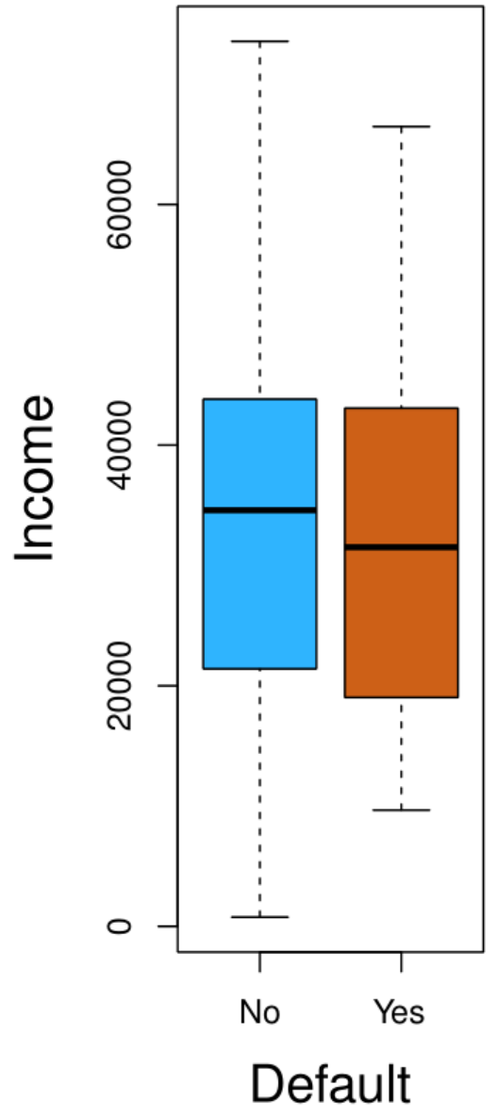


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Example: Credit Card Default







Can we use Linear Regression?

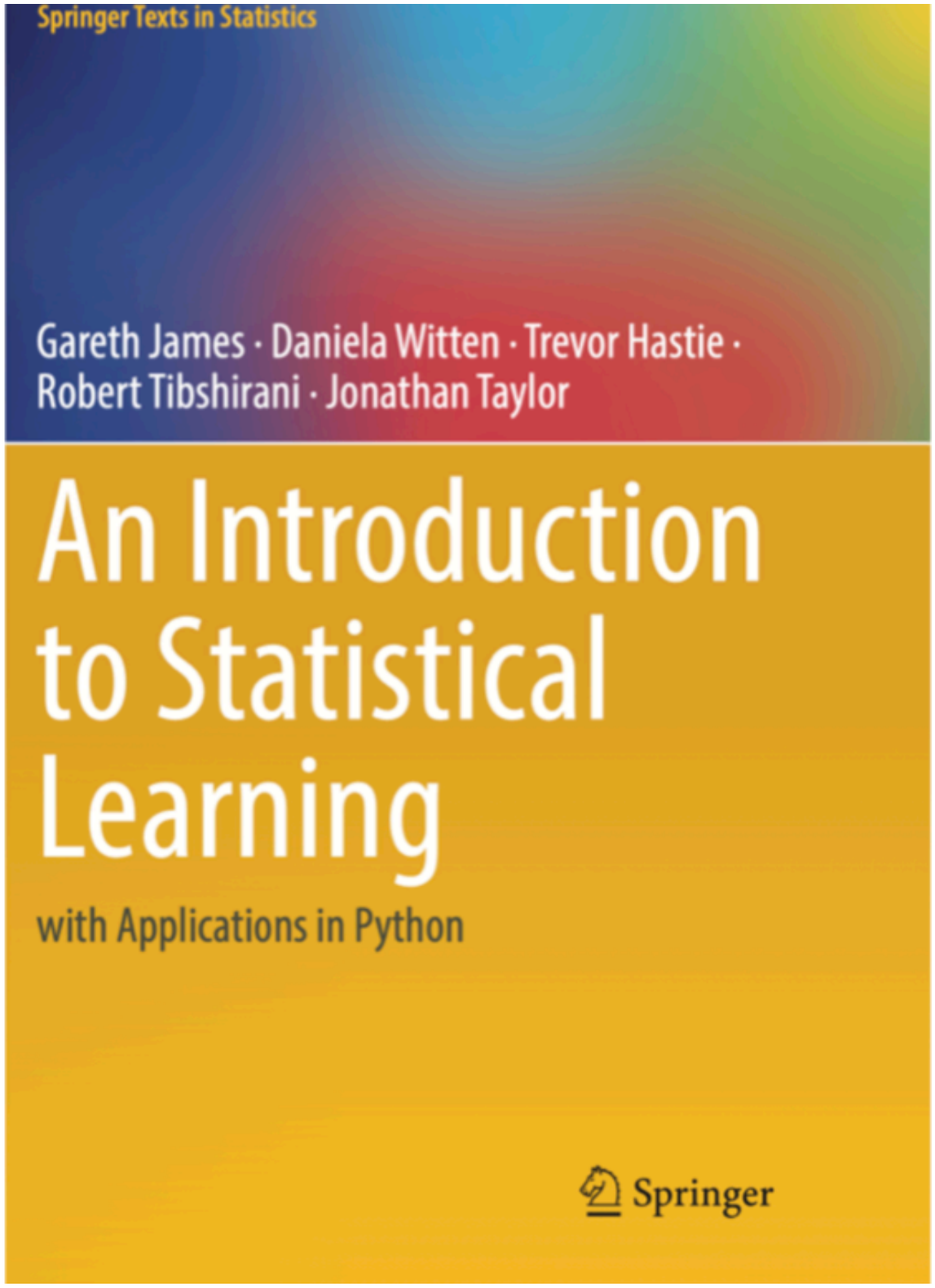
Suppose for the `Default` classification task that we code

$$Y = \begin{cases} 0 & \text{if } \texttt{No} \\ 1 & \text{if } \texttt{Yes}. \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as `Yes` if $\hat{Y} > 0.5$?

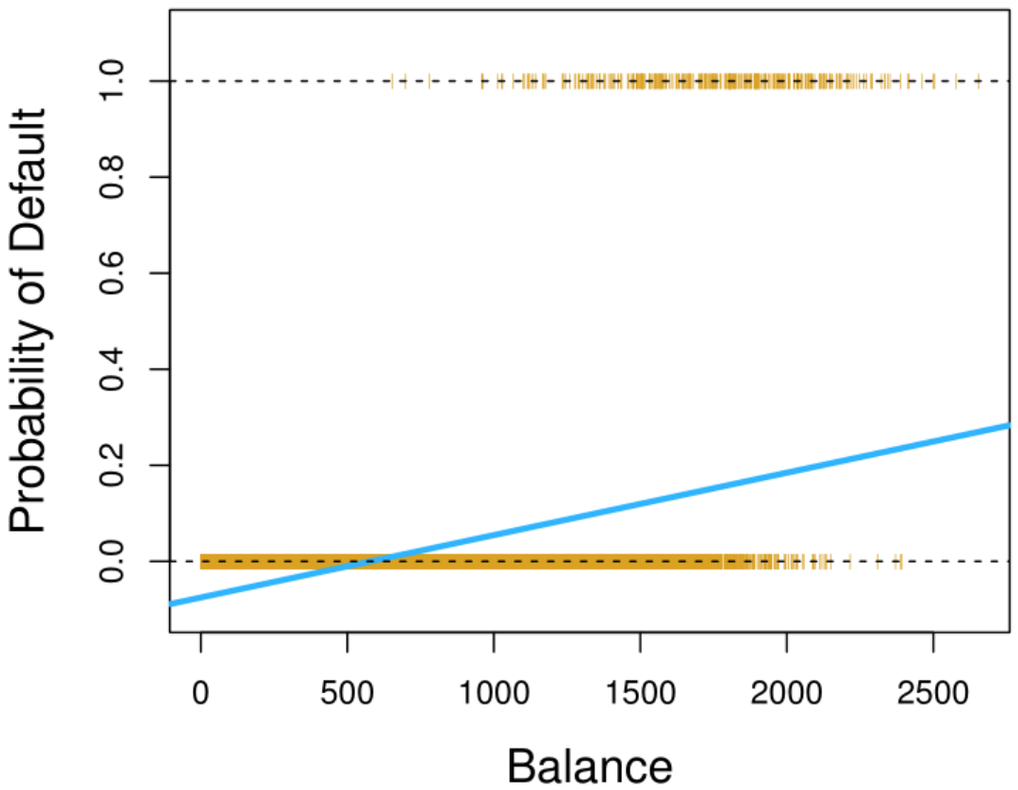
- In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to *linear discriminant analysis* which we discuss later.
- Since in the population $E(Y|X = x) = \Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.
- However, *linear* regression might produce probabilities less than zero or bigger than one. *Logistic regression* is more appropriate.

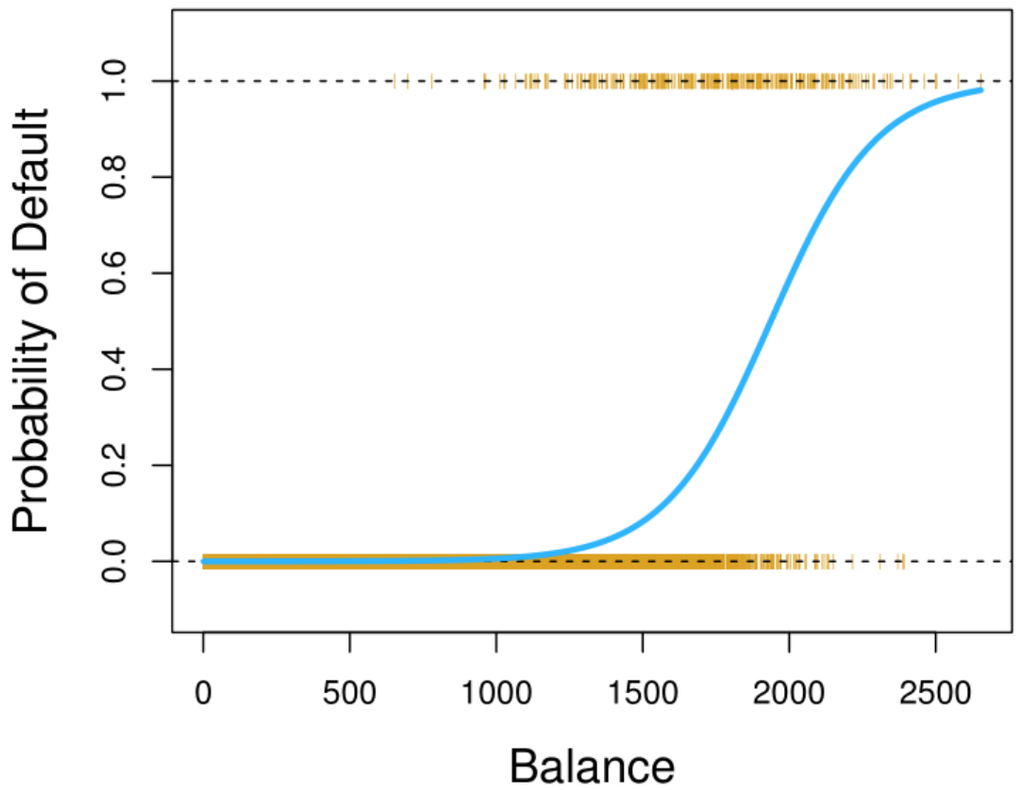
4.3 Logistic Regression



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Linear versus Logistic Regression





The orange marks indicate the response Y, either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Linear Regression continued

Now suppose we have a response variable with three possible values. A patient presents at the emergency room, and we must classify them according to their symptoms.

$$Y = \begin{cases} 1 & \text{if } \texttt{stroke}; \\ 2 & \text{if } \texttt{drug overdose}; \\ 3 & \text{if } \texttt{epileptic seizure}. \end{cases}$$

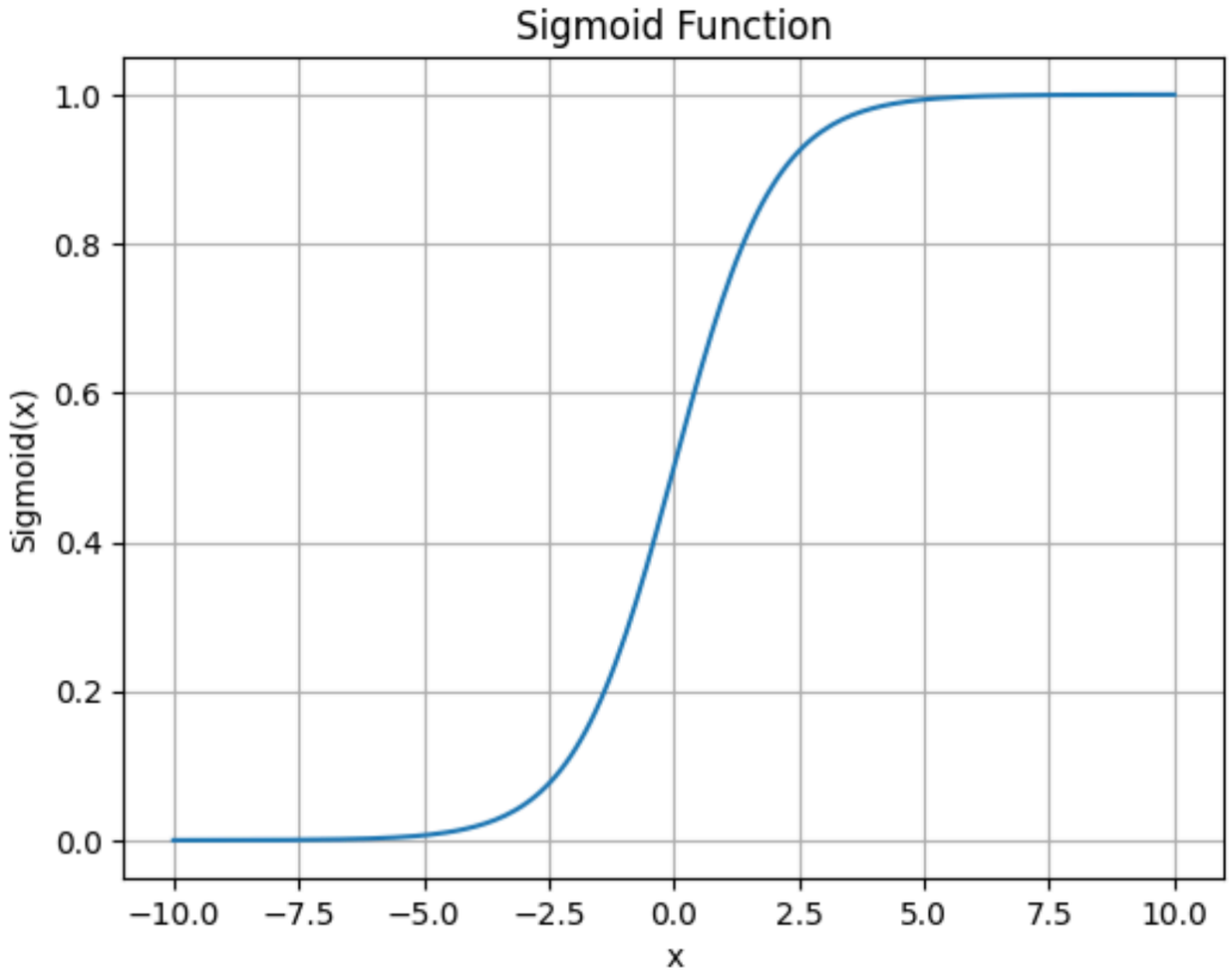
This coding suggests an ordering, and in fact implies that the difference between `stroke` and `drug overdose` is the same as between `drug overdose` and `epileptic seizure`.

Linear regression is not appropriate here.
Multiclass Logistic Regression or *Discriminant Analysis* are more appropriate.

The sigmoid function

How do you draw it by hand?

- $\sigma(t) = \dfrac{1}{1 + e^{-t}}$ for $-\infty < t < \infty$
- Range: bounded in [0,1]
- Shape: S



https://colab.research.google.com/drive/1xnuBEllQrWw3lqb_dpAEuwhmD2ZZfSjd#scrollTo=DjH5fTgmaICM

Linear Regression as a generalized linear model

- Let $\beta = (\beta_0, \beta_1, \ldots, \beta_p)^\top, x = (1, \ldots, x_p)^\top$.
- $y = \beta_0 + \beta_1 x_1 + \cdots \beta_p x_p + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$
- $y = \beta^\top x + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$
- $y \,|\, x \sim N(\beta^\top x, \sigma^2)$
- Error: Normal distribution.
- Mean $\mu(x; \beta) = \beta^\top x$
- $f(y \mid x; \beta) = \dfrac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(y - \beta^\top x)^2}{2\sigma^2}}$

MLE and loss function

- Likelihood

$$L(\beta, \sigma^2) = \prod_{i=1}^{n} f(y_i \mid x_i; \beta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta^\top x_i)^2}{2\sigma^2} \right) = \left(2\pi\sigma^2\right)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta^\top x_i)^2 \right)$$

- Log-Likelihood

$$\log L(\beta, \sigma^2) = -\frac{n}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta^\top x_i)^2$$

- Squared loss: $RSS(\beta) = \sum_{i=1}^{n} (y_i - \beta^\top x_i)^2$

$$-\log L(\beta, \sigma^2) = \frac{n}{2} \log\left(2\pi\sigma^2\right) + \frac{1}{2\sigma^2} \mathrm{RSS}(\beta)$$

$$\Longrightarrow \quad \hat{\beta} = \arg\min_{\beta} \ \mathrm{RSS}(\beta) = \arg\max_{\beta} \ L(\beta, \sigma^2)$$

Logistic Regression as a generalized linear model

- $y \mid x \sim Bernoulli(\beta^\top x)$
  - Error distribution: Bernoulli distribution
  - Mean $\mu(x;\beta) = \sigma(\beta^\top x) = \dfrac{1}{1+e^{-\beta^\top x}} = \dfrac{e^{\beta^\top x}}{1+e^{\beta^\top x}}$
  - $f(y_i \mid x_i;\beta) = \sigma(\beta^\top x_i)^{y_i} \left(1-\sigma(\beta^\top x_i)\right)^{1-y_i}$

Logistic Regression as a generalized linear model

- Likelihood:

$$L(\beta) = \prod_{i=1}^{n} f(y_i \mid x_i; \beta) = \prod_{i=1}^{n} \sigma(\beta^\top x_i)^{y_i} \left(1 - \sigma(\beta^\top x_i)\right)^{1-y_i} = \prod_{i:\, y_i=1} \sigma(\beta^\top x_i) \prod_{i:\, y_i=0} \left(1 - \sigma(\beta^\top x_i)\right)$$

- Log-Likelihood:

$$\log L(\beta) = \sum_{i=1}^{n} \left[y_i \log \sigma(\beta^\top x_i) + (1 - y_i) \log \left(1 - \sigma(\beta^\top x_i)\right) \right]$$

- Binary cross-entropy loss

$$\text{BCE}(\beta) = -\sum_{i=1}^{n} \left[y_i \log \sigma(\beta^\top x_i) + (1 - y_i) \log \left(1 - \sigma(\beta^\top x_i)\right) \right]$$

- $\implies \hat{\beta} = \arg\min_{\beta} \ \text{BCE}(\beta) = \arg\max_{\beta} \ L(\beta)$

Nice properties of $\sigma(t)$

- $\sigma(t) = \dfrac{1}{1+e^{-t}} = \dfrac{e^t}{1+e^t}$
- $\text{odds} = \dfrac{p(Y=1\,|\,x,\beta)}{p(Y=0\,|\,x,\beta)} = \dfrac{\sigma(\beta^\top x)}{1-\sigma(\beta^\top x)} = \dfrac{\frac{e^{\beta^\top x}}{1+e^{\beta^\top x}}}{\frac{1}{1+e^{\beta^\top x}}} = e^{\beta^\top x}$
- Log-odds = $\log(\text{odds}) = \beta^T x$

Probability, odds, log odds, odds ratios

Summary in one line each:

Term	Formula	Range
probability	$p(X) = Pr(Y = 1 \mid X)$	0 to 1
odds	$p(X) / (1 - p(X))$	0 to ∞
log-odds, logit	$\log(p(X) / (1 - p(X))) = \beta_0 + \beta_1 X$	−∞ to ∞
odds ratio	$\text{odds}(X + 1) / \text{odds}(X) = e^{\beta_1}$	0 to ∞

Logistic Regression

Let's write $p(X) = \Pr(Y = 1|X)$ for short and consider using `balance` to predict `default`. Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}.$$

($e \approx 2.71828$ is a mathematical constant [Euler's number.])

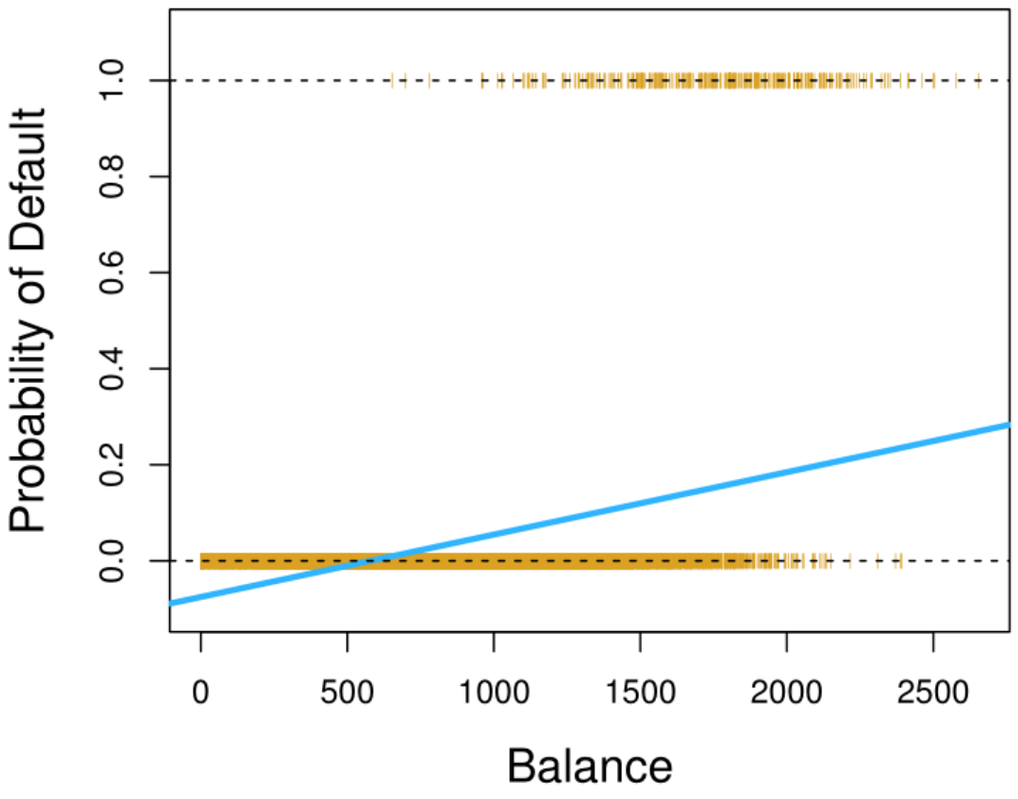
It is easy to see that no matter what values β_0, β_1 or X take, $p(X)$ will have values between 0 and 1.

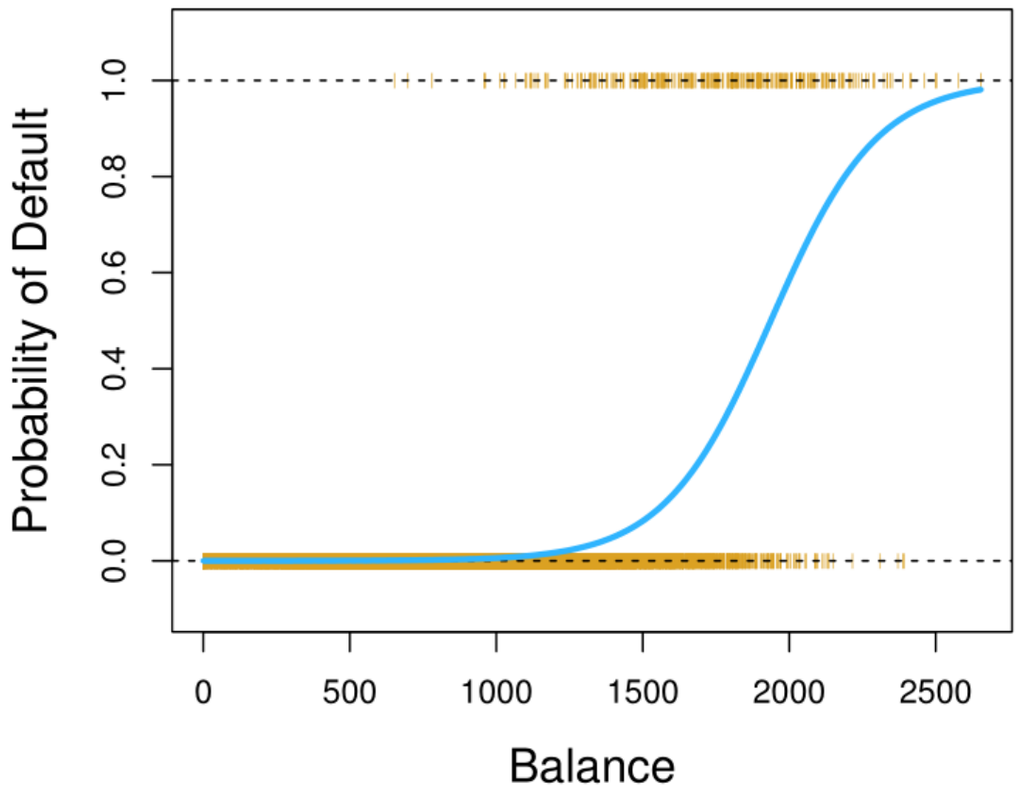
A bit of rearrangement gives

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X.$$

This monotone transformation is called the *log odds* or *logit* transformation of $p(X)$. (by log we mean *natural log*: ln.)

Linear versus Logistic Regression





Logistic regression ensures that our estimate for $p(X)$ lies between 0 and 1.

Maximum Likelihood

We use maximum likelihood to estimate the parameters.

$$\ell(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

This *likelihood* gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.

Most statistical packages can fit linear logistic regression models by maximum likelihood. In `R` we use the `glm` function.

	Coefficient	Std. Error	Z-statistic	P-value
`Intercept`	-10.6513	0.3612	-29.5	< 0.0001
`balance`	0.0055	0.0002	24.9	< 0.0001

Making Predictions

What is our estimated probability of `default` for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1000}}{1+e^{-10.6513+0.0055\times 1000}} = 0.006$$

With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 2000}}{1+e^{-10.6513+0.0055\times 2000}} = 0.586$$

Lets do it again, using `student` as the predictor.

	Coefficient	Std. Error	Z-statistic	P-value
`Intercept`	-3.5041	0.0707	-49.55	< 0.0001
`student[Yes]`	0.4049	0.1150	3.52	0.0004

$$\widehat{\Pr}(\texttt{default=Yes}|\texttt{student=Yes}) = \frac{e^{-3.5041+0.4049\times 1}}{1+e^{-3.5041+0.4049\times 1}} = 0.0431,$$

$$\widehat{\Pr}(\texttt{default=Yes}|\texttt{student=No}) = \frac{e^{-3.5041+0.4049\times 0}}{1+e^{-3.5041+0.4049\times 0}} = 0.0292.$$

Logistic regression with several variables

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

	Coefficient	Std. Error	Z-statistic	P-value
`Intercept`	-10.8690	0.4923	-22.08	< 0.0001
`balance`	0.0057	0.0002	24.74	< 0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	-0.6468	0.2362	-2.74	0.0062

Why is coefficient for `student` negative, while it was positive before?